

Limitations

- Scope limited to single site
- No Handling of distributed transactions
- No n/w or communication failure management.
- Limited recovery scope
- No Global consistency enforcement.

6. (a) Soln

The main parallel DBMS techniques used to improve query performance are:-

1. Data partitioning: Splitting data ^{base} into multiple partitions on multiple nodes.
2. Intra query parallelism: parallel execution of different query parts.
3. Inter-query parallelism: parallel execution of multiple operations.
4. Pipelining: Intermediate result of one of the operation passed to another without ~~the~~ waiting for full operation.

6.

Data partitioning

→ Splitting large database into multiple partitions on multiple nodes.

Types: -

- 1. Horizontal partitioning
 - 2. Vertical partitioning
 - 3. Range partitioning
 - 4. List partitioning
 - 5. Index partitioning
- } not needed

Problem lead with partitioning in parallel execution.

Data skew (Uneven Data Distribution)

Data is unevenly distributed across the sites.

for example, A sales data at node A are 90000 and at node B are 10000.

Replication overhead

Replication of large schema can lead storage and communication cost.

partitioning Attribute not aligned with query.

possible solution:-

for skew data, we can use hash partitioning.

for example: for above exo

- for given example, we can use hash function on sales ID.

$$\text{Partition} = h(\text{sales ID}) \bmod N$$

Similarly,

problem

proposed solution,

ii) Replication overhead

→ partial replication or selective replication,

iii) Mismatch partition attributes

→ Repartition on join

problem lead with pipelining.

1. Blocking operators

→ operators like GROUP BY, SORT, DISTINCT, must process all inputs before producing any output, this blocks pipeline.

- for example: find avg salary per department for

1. Department, Avg(salary) (Employee).

- Here group by need to see all rows for each department to calculate average salary.

This mean it has to wait until entire process.

Other problems

→ Data dependency delay: one of the operation depend on other for output.

a)

→ Communication overhead: Difficult to communicate output of one as input of another.

Proposed solution

1. ~~for~~ solution of blocking operation is hash based streaming.

for given example, Instead of waiting use hash function as,

- $h(\text{Department})$, send all rows to same processor.

- for upcoming row, the processor updates sum.

→ finally calculate averages

~~also~~ similarly,

Data dependency delay → Solution
Use Incremental partition.

Communication overhead → Compression, Join with keys.

6. (b) soln

The role of cost based optimization in distributed query processing are:-

i) Estimate costs.

→ Estimate costs for scanning, joining and transferring data at each site.

ii) Generate Execution plans.

→ Generate multiple query plans with different orders of operations.

iii) Evaluate plans:

→ Calculate cost of each plan based on CPU, I/O and data transfer.

iv) Select Best plan.

→ choose best plan with minimum cost

Let's understand through example.

Relation $R(A, B)$ stored at site 1; size $|R| = 1000$ tuples

By,

Relation $S(B, C)$ at site 2; size $|S| = 50000$

Query: $Q \sim R \bowtie_{R.B=S.B} S$

STEP 1:

Let, network cost per tuple: $C_{net} = 5$ unit

Local CPU/I/O cost per tuple: $C_{local} = 1$ unit

Step 2: Generate Execution plans.

Plan	Description
1.	Ship entire R to site 2, joint at site 2
2.	Ship entire S to site 1, joint at site 1

Step 3: Evaluate plans.

Estimate total cost = Cost_{transfer} + Cost_{local}.

Plan 1: Ship R to site 2.

Transfer cost:

$$|R| \times C_{net} = 1000 \times 5 = 5000$$

Local join processing cost at site 2.

$$(|R| + |S|) \times C_{local} = (1000 + 5000) \times 1 = 6000$$

$$\therefore \text{Total cost} = 5000 + 6000 = 11000 \text{ unit}$$

Plan 2: Ship S to site 1.

Transfer cost

$$|S| \times C_{net} = 5000 \times 5 = 25000$$

Local join processing cost at site 1

$$(|R| + |S|) \times C_{local} = (1000 + 5000) \times 1 = 6000$$

$$\therefore \text{Total cost} = 25000 + 6000 = 31000$$

Step 4: Selection of best plan

$\therefore$ Ship entire Q to site 2, join at site 2 has minimum cost. Thus better selection.